

ABHIJITH SIVAPRASADAN

Thermal-Fluid & Energy-Systems Researcher | CFD & Heat Transfer, Thermal Energy Storage, Dynamic/Digital-Twin Modelling & Energy-Systems Optimisation

Solna, Sweden | +46 76 969 2014 | abhijithsivaprasadan@gmail.com
[linkedin.com/in/abhijith-sivaprasadan](https://www.linkedin.com/in/abhijith-sivaprasadan) | github.com/abhijith-sivaprasadan | abhijith-sivaprasadan.github.io

PROFILE

Thermal-fluid and energy-systems researcher with an M.Sc. in Sustainable Energy Engineering from KTH Royal Institute of Technology and a B.Tech in Mechanical Engineering. My research background spans CFD and conjugate heat-transfer modelling of high-temperature industrial hardware (master thesis at Siemens Energy's Fluid Dynamics Laboratory), laboratory characterisation of thermal energy storage and HVAC/refrigeration systems, dynamic and digital-twin process simulation, and energy-systems optimisation (MILP) applied to low-carbon power, hydrogen and grid-flexibility problems. Independent cross-checking, through an analytical calculation, a second implementation, automated tests or measured data, is a recurring part of my research and modelling workflow. **Research interests:** high-temperature thermal-fluid systems and heat transfer; thermal energy storage charge-discharge dynamics; dynamic and digital-twin process/plant modelling; energy-systems optimisation and grid flexibility; data-driven and reduced-order modelling; model verification methodology.

EDUCATION

- M.Sc. Sustainable Energy Engineering** Aug 2023 - 2026
KTH Royal Institute of Technology, Stockholm
 - 123 ECTS (120 required + 3 elective) complete per official KTH confirmation letter (24 Sep 2026); degree certificate/conferral in progress. 30 ECTS thesis passed May 2026 (TRITA-ITM-EX 2026:14).
- B.Tech Mechanical Engineering** 2017 - 2021
College of Engineering Perumon, APJ Abdul Kalam Technological University, India

FULL GRADUATE COURSEWORK

Code	Course	ECTS	Grade
MJ232X	Degree Project in Heat and Power Technology (master thesis)	30	Pass
MJ2515	Numerical Heat Transfer in Energy Technology	3	A
MJ2507	AI Applications in Sustainable Energy Engineering	6	A
MJ2386	Energy Storage Technology	6	B
MJ2411	Renewable Energy Technology	6	B
MJ2426	Applied Heat and Power Technology	6	C
MJ2510	Methods of Research in Sustainable Energy	7.5	C
MJ2511	Energy Management	7.5	C
ME2072	Entrepreneurship for Engineers	6	C
MJ2503	Small Scale Polygeneration	6	C
MJ2508	Energy Systems for Sustainable Development	6	C
MJ2505	Practical Optimization of Energy Networks	6	D
MJ2405	Sustainable Power Generation	9	D
MJ2438	Modeling of Energy Systems – Heat and Power Generation	6	D
MJ2509	Energy in the Built Environment	9	E
—	Circular Economy for Energy Storage Theory D (Aalto/Unite!, credited)	3	—

RESEARCH AND PROFESSIONAL EXPERIENCE

- Master Thesis Student, Fluid Dynamics Laboratory** Apr - Nov 2025
Siemens Energy AB, Finspång | Numerical Investigation of Steady-State Thermo-Fluid Performance of a Reducer for a High-Temperature Dynamic Pressure Sensor Calibration Rig (TRITA-ITM-EX 2026:14)
 - Built compressible CFD and conjugate heat-transfer models in ANSYS Fluent (SST k- ω) of high-temperature industrial hardware, verifying spatial-discretisation sensitivity with a three-level mesh-independence study; an analytical heat-loss consistency check agreed with Fluent within 0.1% across six CHT cases.
 - Commissioned the measurement chain for the planned experimental validation campaign using NI-DAQ/NI MAX, LabVIEW and thermocouple/pressure channels; when an internal heater failure occurred during experimental preparation, before the high-temperature campaign began, conducted the root-cause analysis with the industrial supervisor and delivered a re-scoped 15-run numerical study within the original schedule.
- Energy Efficiency Intern** Dec 2024 - Jun 2025
Alleima AB (remote)
 - Developed a comparative industrial energy-audit methodology across six countries, including a tiered ISO 50001/EED-oriented framework, KPI/EnPI baselines and load-driver normalisation.
- Engineer (backend software)** Sep 2021 - Jun 2023
QBurst Technologies, Thiruvananthapuram, India
 - Developed production backend services (Go, later TypeScript/NestJS) with automated tests and negative-path validation; underpins the reproducibility and verification discipline applied across all research/modelling work below.

ADDITIONAL RESEARCH EXPOSURE

Student Research Project / Research Engagement – Pyrolysis

Feb - Jun 2024

Division of Process Technology, KTH (with Akhil Mohan, Post-Doctoral Researcher)

- Informal research engagement reviewing waste-polymer pyrolysis reactor/process concepts and techno-economic reasoning for a Waste-to-Wealth research project.

THERMAL ENERGY STORAGE & HEAT TRANSFER PROJECTS

M-TES PCM discharge characterisation laboratory

KTH laboratory (MJ2386), group project

- Characterised the discharge of a 44 L/33 kg RT60 HC latent-heat storage prototype using ten PT100 sensors, a Type-T thermocouple and a Keithley DAQ over an 8-hour test; theoretical $Q_{\text{storage}} = 10,049.2$ kJ vs. measured Q_{HTF} (Method 1) = 10,035.6 kJ (0.14% agreement), with the storage tank's own thermal mass identified as the dominant unmodelled term through residual analysis.

TES Discharge Screen: thermal-storage technology screening framework

Python, Pyomo/HiGHS, Modelica/FMI

- Built a packed-bed dynamic thermal model alongside molten-salt and PCM sub-models, cross-checked a Modelica FMU (OpenModelica/FMPy) against an independent Python model over an 8-hour test (0.23% maximum temperature deviation); a software reference study, not validated against a real plant.

R134a refrigeration / heat-pump performance and AHU psychrometrics laboratories

KTH (MJ2509)

- Evaluated refrigeration/heat-pump operation across three evaporation conditions (COP, pressure-enthalpy analysis) and cooling-coil performance at three airflow conditions (wet-dry-bulb measurements, psychrometric-chart analysis), including identifying and reporting an inconsistent data point as likely measurement error.

Numerical methods for heat-transfer and transport problems

KTH (MJ2515), Grade A, Python

- Implemented and compared numerical methods for heat-transfer and transport problems, including 1-D/2-D finite-difference conduction, transient time integration (Euler/RK4) and advection schemes; analysed discretisation error, residual convergence and stability against analytic/reference solutions.

DYNAMIC & DIGITAL-TWIN MODELLING PROJECTS

ThermoTwin-F: gas-turbine, combined-cycle and grid-operation simulator

Modern Fortran (Fortran 2008), independent project

- Built an inspectable simulator connecting component thermodynamics to plant operation: design-point performance, degradation/uncertainty studies, synthetic-fault inference, off-design/part-load behaviour, combined-cycle/HRSG and grid-dispatch/frequency-response modules; verified with an automated test suite and a reproducible, hash-logged analysis pipeline. Framed per its own documentation as a research/educational model, not validated against a commercial engine or plant.

OpenSteamOpt: steam-and-power optimisation and Modelica digital-twin demonstrator

Python, Pyomo/HiGHS, Modelica/FMI

- Built a two-boiler/two-header steam-and-power plant demonstrator coupling MILP dispatch optimisation with a Modelica dynamic plant model, verified via FMI 2.0 co-simulation and cross-checked against an independent Python shadow-twin model; synthetic scenarios, not real-plant savings.

Non-Gray Thermal Radiation Modeling

Python, independent project

- Built a non-gray thermal-radiation modelling toolkit around published combustion/fire datasets: a transparent 1-D discrete-ordinates solver, Mie-correlation checks and surrogate modelling with reproducible visualisation and reporting workflows.

Structural FEA: reactor-internals-inspired support plate

ANSYS Mechanical/SpaceClaim, independent project

- Ran static and combined-loading structural analysis with a mesh-convergence study and a thermal-expansion sensitivity sweep (up to +100°C) on a representative support-plate geometry; an educational screening study, not ASME BPVC or certified nuclear design.

ENERGY-SYSTEMS OPTIMISATION PROJECTS

PyNEXUS: electricity-hydrogen-heat optimisation model

Python, Pyomo/HiGHS, independent project

- Built a coupled offshore-wind, PEM-electrolyser and hydrogen-pipeline MILP with optional storage, grid and heat-recovery coupling, with a 168-hour reference configuration and a recorded 8,760-hour annual run using real ERA5 wind data, plus independent dispatch verification and reconciliation checks.

PyPSA-NL: exploratory grid-flexibility model

Python, PyPSA/Linopy/HiGHS, independent project

- Built a Netherlands-inspired grid-flexibility screening model covering regional buses/transmission, renewables/backup/BESS siting and sizing sweeps, flexible-connection logic and an N-1 screening proxy; exploratory, not validated TSO-grade planning.

Energy KPI Toolkit and EU ETS Exposure Calculator

Python, Excel/Streamlit, independent projects

- Developed an industrial energy-KPI toolkit for EnPI baselining, production/operating-hour normalisation, anomaly/drift screening and automated management reporting, and a companion Streamlit/Excel tool for Scope 1/2 emissions accounting, EU ETS EUA-exposure scenarios and carbon-intensity reporting (synthetic scenario tools, not live compliance analysis).

Day-ahead machine-learning forecast of building heating demand

KTH (MJ2507), Grade A,
Python, group project

- Helped build a day-ahead heating-demand forecasting workflow for a real, approximately 3,600 m² KTH university building: multivariate data cleaning, Mahalanobis-distance outlier filtering and feature scaling, comparing a baseline model against shallow and deep neural networks on held-out operation data.

Independent software projects above use AI-assisted implementation; I define the problem formulation, model architecture, validation criteria and interpretation of results.

LEADERSHIP & EXTRACURRICULAR

Process Manager / Wardrobe Manager / ETF Head / Guest & Safety

Sep - Dec 2023

Students' Nobel NightCap, Stockholm

- Coordinated guest, safety and event-logistics processes across operational teams for a high-visibility student-led event under tight timelines and shared operational responsibilities.

Treasurer, Secretary & Team Captain (SAE Bicycle Design Competition)

2019 - 2021

SAE Student Chapter, College of Engineering Perumon

- Treasurer (2019-20) and Secretary (2020-21) of the student chapter; Team Captain of the SAE Bicycle Design Competition 2020; also contributed to BAJA/tractor-design team projects.

Automatic Sanitiser Dispenser – project lead

2020

College of Engineering Perumon (NSS-backed campus initiative)

- Led an Arduino ultrasonic/servo-actuated sanitiser dispenser project from concept through a compact SolidWorks redesign, building the first unit from e-waste/PC-chassis components and supporting small-batch local deployment with faculty/NSS backing.

TECHNICAL SKILLS

CFD / FEA

ANSYS Fluent (compressible flow, CHT, SST $k-\omega$); ANSYS Mechanical (static/combined loading, mesh convergence); mesh-independence methodology

Thermal systems & instrumentation

Thermal energy storage, refrigeration/HVAC cycles, psychrometrics; PT100/thermocouple sensors, Keithley/Ni-DAQ

Dynamic & digital-twin modelling

Modern Fortran, Modelica/OpenModelica, FMI 2.0 co-simulation; model-vs-model and model-vs-measurement verification

Energy-systems optimisation

Pyomo/HiGHS and PyPSA/Linopy MILP/LP formulation; KPI/EnPI and emissions-accounting modelling

Data-driven modelling

Python (pandas, NumPy, scikit-learn), multivariate data cleaning/outlier detection, shallow/deep neural networks

Programming

Python, MATLAB, Modern Fortran, Go, TypeScript; Git, reproducible research workflows, automated testing

ADDITIONAL INFORMATION

Master's thesis / Research output: TRITA-ITM-EX 2026:14 – Numerical Investigation of Steady-State Thermo-Fluid Performance of a Reducer for a High-Temperature Dynamic Pressure Sensor Calibration Rig (KTH master thesis, 2026).

Conference: Sponsored student delegate, Converging Energy Technologies (CET2026), Oskarshamn, Sweden, 28 Sep - 1 Oct 2026

Languages: English (professional working proficiency), Malayalam (native), Swedish (actively studying)